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Research Paper

Amendments of nitrogen and sulfur mitigate carbon-promoting effect on microbial mercury methylation in paddy soils

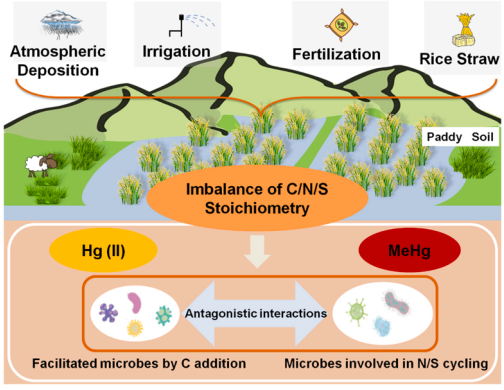
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HIGHLIGHTS

- Imbalanced nutrient elements input affects MeHg production in paddies.
- The addition of N or S mitigates the C-promoting effects on MeHg production.
- Changes in microbiome induced by altered C/N/S ratios affect net MeHg production.
- Microbial response mechanism for MeHg production linking to C/N/S ratio was discussed.

GRAPHICAL ABSTRACT



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ABSTRACT

The imbalance of nutrient elements in paddy soil could affect biogeochemical processes; however, how the key elements input influence microbially-driven conversion of mercury (Hg) to neurotoxic methylmercury (MeHg) remains virtually unknown. Herein, we conducted a series of microcosm experiments to explore the effects of certain species of carbon (C), nitrogen (N) and sulfur (S) on microbial MeHg production in two typical paddy soils (yellow and black soil). Results showed that the addition of C alone into the soils increased MeHg production approximately 2–13 times in the yellow and black soils; while the combined addition of N and C mitigated the C- promoting effect significantly. Added S also had a buffering effect on C-facilitated MeHg production in the yellow soil despite the extent being lower than that of N addition, whereas this effect was not obvious for the black soil. MeHg production was positively correlated with the abundance of Deltaproteobactera-

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hgcA in both soils, and the changes in MeHg production were related to the shifts of Hg methylating community resulting from C, N, and S imbalance. We further found that the changes in the proportions of dominant Hg methylators such as *Geobacter* and some unclassified groups could contribute to the variations in MeHg production under different treatments. Moreover, the enhanced microbial syntrophy with adding N and S might contribute to the reduced C-promoting effect on MeHg production. This study has important implications for better understanding of microbes-driven Hg conversion in paddies and wetlands with nutrient elements input.

1. Introduction

The inorganic mercury (Hg) in aquatic environments can be converted to neurotoxic methylmercury (MeHg) that easily bioaccumulates and biomagnifies along food webs, posing risks to ecology security and human health [15,33,49]. Mercury methylation is primarily driven by certain microbes carrying *hgcAB* genes such as sulfate-reducing bacterial (SRB), iron-reducing bacterial (FeRB), and methanogens, as well as some newly discovered taxa like Asgard archaea from metagenomic analyses [13,30,42,43,63]. Recently, the consumption of rice grain has been considered another important pathway of MeHg exposure to specific populations in addition to the well-known seafood [21,65,67], and the MeHg in rice mainly comes from the uptake and transportation of soil MeHg [39,64]. Water-saturated paddy during rice growth is a hotspot for MeHg production because of the enrichment of anaerobic Hg methylators (i.e., *Methanogens* and *Geobacterales*) [33,68]. Exploring the factors and underlying mechanisms affecting MeHg production in paddy soils is critical for mitigating MeHg accumulation in rice grain, as well as the security of downstream aquatic environments.

Agricultural practices such as fertilization and irrigation can lead to an imbalance of nutrient elements in paddy soils. Among them, carbon (C), nitrogen (N), and sulfur (S) are three important elements for croplands and play pivotal roles in maintaining soil fertility and plant growth [2,25,40]. However, changes in the soil stoichiometry of C/N/S resulting from anthropogenic activities can substantially affect the biogeochemical processes, including the transformation of nutrients and pollutants [11,27]. Many studies had demonstrated that soil conditions like the supply of nutrient elements markedly affected the production of MeHg [24,26,28,41,66]. Earlier studies showed that anaerobic microbes such as SRB, FeRB, and methanogens in soil were susceptible to the addition of C, N, and S [17,18,4]. For example, nitrate fertilizer application induced an imbalance of C/N, and thus affected the activities of SRB and methanogens [17,38,9]. In addition, the enhancement of dissolved organic carbon increased the percentages of available C and N and the relative abundances of *Methanoregula* and *Geobacter* in paddy soils (Shanghai and Hunan Province in China) [57,62]. Therefore, soil C/N ratio is important for Hg methylation in environments [31,34,53]. Moreover, recent studies showed that sulfur fertilizer amended in paddy soils resulted in increases in the relative abundance of *Desulfomonile tiedje* and *Desulfovibrio spp* (Hg methylators) [60]. Similar to nitrate, sulfate can also act as an inhibitor to compete for substrate with methanogens [44]. Therefore, we expected that the stoichiometric ratios of C/N/S are closely associated with the production of MeHg in paddy fields, which are driven by certain anaerobic microorganisms. However, it is unknown how the imbalanced input of C, N, and S may influence the microbially-driven MeHg production in paddy soils.

Here, we hypothesize that certain species and levels of C, N, and S may influence microbial Hg methylation in paddy soils. To test our hypothesis, we conducted a series of microcosm experiments to investigate the effects of C, N, and S input at real-world levels on MeHg production in two typical paddy soils (yellow and black soils). We sought dynamics of MeHg accumulation within 50 days under different C, N, and S treatments. We then characterized Hg methylating communities by quantifying and sequencing *hgcA* genes at the end of incubation. We also analyzed the overall bacterial community through sequencing of *16 S rRNA* gene amplicons. The results based on these combined analyses may provide new insights into Hg methylation in paddy soils responding to changes in nutrient elements input under intensified agriculture for ever-increasing food demands.

2. Materials and methods

2.1. Sample collection and microcosm setup

The yellow and black paddy soils (0–20 cm) were collected from two typical rice planting areas, respectively, Guangdong Province (23°16'N, 113°36'E) in South China and Heilongjiang Province (47°27 N, 126°55 E) in North China, which have distinct climatic conditions. Guangdong Province is located in the subtropical monsoon climate zone, with an average annual temperature of 22.0 °C and an average annual precipitation of 1738 mm. Heilongjiang Province is located in the temperate monsoon climate zone, with an average annual temperature of 1.5 °C and an average annual precipitation of 550 mm. Different climate and geographical conditions determine the distinct soil properties. The collected surface soils were transported to the laboratory in ice chambers, ground, and sieved through 2 mm sieve to remove stones and plant residues, and then stored at 4 °C for further incubation experiments. The chemical properties of these two typical soils were shown in Table S1.

A series of experiments were first designed to investigate the effects of C, N, and S input on MeHg production. Experimental soil (equivalent to 100 g of oven-dried weight) was placed in a polyvinylchloride container (7.5 cm diameter × 10 cm height) and then pre-incubated at 25 °C in darkness for one week before Hg methylation assays. The levels of C (glucose), N (sodium nitrate), S (sodium sulfate) and Hg(II) ($HgCl_2$) were 500 mg kg^{-1} (C/soil), 250 mg kg^{-1} (N/soil), 45 mg kg^{-1} (S/soil) and 1 mg kg^{-1} (Hg/soil, lower than that in the soils of rivers [19]), respectively, simulating elements input in the fields. C, N, S and Hg were added in the form of water solution. The soil: deionized water ratio was about 1:5 in the incubation systems. Seven treatments were set to simulate changes of stoichiometric ratios of C/N/S in paddy soils, including Hg, Hg+C, Hg+N, Hg+S, Hg+CN, Hg+CS, Hg+CNS treated groups (Table S2). A layer of three centimeters level of water was kept above the soil surface to maintain the anaerobic environment during the incubation experiment. Each sealed treatment was in triplicate and incubated at 25 °C for 50 d. The lower soil was collected at each sampling time point (0, 7, 14, 28, and 50 day) and freeze-dried after centrifugation (3000 r/min, 15 min, and room temperature). The MeHg production in all treatments was timely monitored. Based on the dynamics of MeHg production, incubated soils at 50 days were selected to analyze the total bacterial and Hg methylating microbial communities.

2.2. Soil chemical analyses

Soil Eh and pH (soil / water = 1/2.5, m/m) were measured in situ using a portable HQ40d multi-parameter meter (Hach, USA). Total C, N, and S were determined by elemental analyzer (Elementar vario macro cube, Germany). NH_4^+ and nitrate in the soil was extracted with 2 M KCl solution and determined through continuous flow analyzer (AA3, Germany SEAL) after filtration. Sulfate and DOC in the soil was determined through ion chromatograph (Metrohm 930, Switzerland) and TOC analyzer (Multi C/N 2100, Germany) after extracting by ultrapure water. Approximately 100 mg freeze-dried samples through 0.149-mm sieve were weight and placed in the centrifuge tube (15 mL, Corning), and then leached with 7.5 mL nitric acid (25%, v/v) and 1.5 mL $CuSO_4$

(2 mol L^{-1}). Afterwards, MeHg in the supernatant was extracted with methylene chloride and water and determined by cold vapor atomic fluorescence spectrometry (CVAFS, Brooks Rand Model III, USA) with the detection limit of 0.002 ng L^{-1} for MeHg. The spiked recoveries of the reference material (ERM-CC580) ranged from 88.7% to 110.0%, and the values of analytical blank for MeHg were lower than the detection limit (0.002 ng L^{-1}).

2.3. Microbial community analyses

Soil samples at 50 d were chosen for total DNA extraction, quantification of *hgcA* gene and amplicon sequencing analyses in order to explore the effect of C/N/S stoichiometric ratios on the microbial communities associated with MeHg production. Total DNA was extracted from 0.25 g freeze-dried soil using a Mo Bio Power Soil DNA isolation kit (Mo Bio Laboratories, Inc. Carlsbad, CA, USA) following the instructions of the manufacturer. Soil DNA extract was subjected to a real-time quantitative PCR (qPCR) to determine *hgcA* copy number [16,33]. The thermal cycling parameters were shown in Table S3, and the qPCR results were accepted when amplification efficiencies were in the range between 85% and 110%. Amplicons were sequenced using primer pair *hgcA*_261F/*hgcA*_912R with barcode adaptors by the Illumina MiSeq (PE 300) at Majorbio in Shanghai, China. Only the forward read sequence was used for downstream analysis due to the fragment length of the *hgcA* gene exceeding the available sequencing length. Original sequences of *hgcA* amplicon sequencing were analyzed using USEARCH version 8.0. Hidden Markov Model was used to search the original sequence of *hgcA* amplicon sequencing for taxonomical annotation with a manually curated database of Hg-MATE-Db.v1.01142021 using HMMER [14].

The qPCR of soil functional genes including *N arG* , *NirS* and *NirK* was performed on an iQ5 thermocycler (Bio-Rad, USA) in a 25-μL reaction mixture containing 12.5 μL ChamQ Universal SYBR qPCR Master Mix (Vazyme , Naning, China). After amplification, a standard curve between the plasmid fluorescence value and copy number was used to calculate the copy numbers of each sample gene. Primer sequence and amplification procedure of functional genes were shown in Table S3.

2.4. Statistical analysis

One-way ANOVA and LSD with multiple comparisons were conducted to determine differences ($p < 0.05$) among different treatments. Random forest analysis was performed to identify microbial predictors of MeHg production [3]. Increases in the percentage of mean squared error (MSE%) were calculated to estimate the importance of factors contributing to the distribution of MeHg using the "rfPermute" package in R 4.0.2 (http://cran.r-project.org/). Greater values of MSE% indicate a greater importance of variables. Finally, we evaluated correlations between the MeHg production and the relative abundances of selected Hg methylators and non-Hg methylators.

3. Results and discussion

3.1. Soil MeHg production responding to C N and S input

C, N and S, as three important elements in croplands, can significantly affect soil chemical properties, thus affecting the bioavailability of soil Hg. For example, previous studies showed that adding S increased dissolved Fe concentrations and further decreased soil Eh [36,56]. Nitrate addition improved Eh value due to the oxidation of reduced sulfur and oxidation of other electron donors by nitrate [58]. The presence of organic matter can improve soil reduction intensity because of the oxidation of organic matter and mobilization of anaerobic microorganisms [46]. In the present study, it showed that exogenous C, N, or S inputs all enhanced the soil MeHg production during the whole incubation period (0–50 d) in both the yellow and black soil (Fig. 1A-B; Fig. S1A-B). The results could be attributed to the increased proportions and activity of Hg methylators by nutrient inputs into the soils [32,38,60,62]. For example, previous studies indicated that agricultural straw return to fields promoted MeHg production in paddy soil because of the increased dissolved organic carbon and stimulated microbial activities [32,52]. The net production of MeHg increases with dissolved organic carbon to total Hg ratio [22]. However, we found that combined C and S amendment significantly mitigated the enhancement of MeHg production in the yellow soil during 50 day's incubation compared to that with

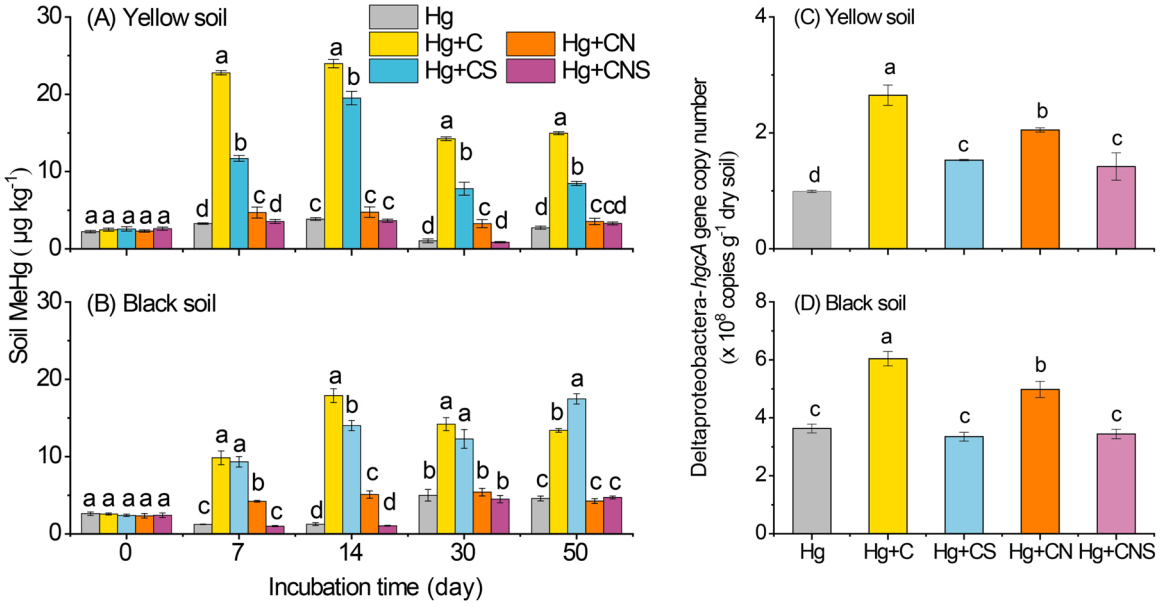


Fig. 1. The effects of stoichiometric ratio of C/N/S on microbial mercury (Hg) production. (A, B) Differences in MeHg production in the yellow and black soil during the incubation period. (C, D) Deltaproteobactera-*hgcA* abundance under the different treatment at the end of incubation in the two soils. Hg, Hg+C, Hg+CS, Hg+CN and Hg+CNS represent Hg-spiked soil amended with different ratios of carbon (C), nitrogen (N) and sulfate (S) as described in the Materials and methods. Different small letters above the columns indicate significant differences among different treatments at the same incubation time ($p < 0.05$). The error bars represent the SD of the means (n = 3).

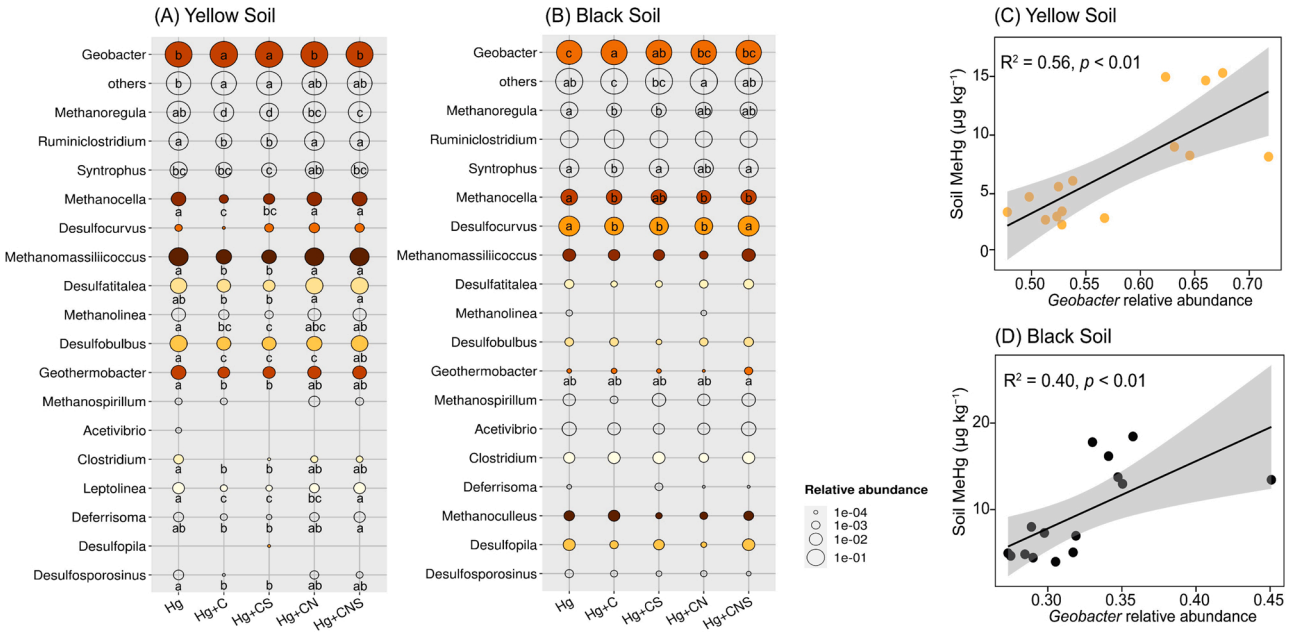


Fig. 2. The effects of stoichiometric ratio of C/N/S on community composition of Hg methylators. (A, B) Community composition of Hg methylators at 50 d of Hg methylation assays in the yellow and black soil; (C and D) Relationships between soil MeHg and *Geobacter* of relative abundance in the two soils. Different small letters inside the cycle indicate significant differences among different treatments ($p < 0.05$). The error bars represent the SD of the means (n = 3).

single C treatment, suggesting that C/S ratio is an important regulator of Hg methylation. In comparison to the control group (without C or S input), the MeHg production increased by 5.21 (at 7 day) to 12.64 (at 30 day) times with single C inputs, and by 2.58 (at 7 day) to 6.46 (at 30 day) times with combined C and S inputs in the yellow soil. Redox variations could influence the biogeochemical behaviors of inorganic Hg and MeHg indirectly through surrounding environment changes induced by dissolved organic carbon, S cycle and dissolved Fe^{3+}/Fe^{2+} [22]. Combined input of C and S increased the soil Eh from 0 to 14 d, which is a potential reason for the declined C- promotional effect on MeHg production (Fig. S2A) [36,56]. However, this buffering effect of S addition disappeared for the black soil after 30 days incubation, which could be related to the relatively low level of sulfate in the soil (Table S1). These results could be explained that high level of sulfate in the soil is prone to be utilized by other anaerobic microbes rather than only Hg methylators, and the competition for resources may result in a declined activity of microbial Hg methylation [29]. These observations are in line with a previous study showing an inhibitory effect of sulfate amendment on Hg methylation in freshwater sediments [48].

Importantly, our results showed that N addition consistently weakened the promoting effect of C amendment on MeHg production in both soils (Fig. 1A-B). Similar phenomena also occurred in the combined C, N, and S treatment with a low C/N ratio during the incubation (Fig. 1A-B). Lower MeHg production under low C/N condition could be attributed to reduced microbial activities (e.g., SRB) in the presence of high level of nitrate [51]. The changes may limit the proliferation and activity of Hg methylating microbes due to the strong competition for electron donors (e.g., lactate), for instance, the competition between sulfate and nitrate reducing bacteria [12,66]. In the present study, we found that N addition increased NH_4^+ concentration in the yellow soil with increasing incubation time (Fig. S3A), while C or S input had no effect on NH_4^+ concentration in the soil. These results imply that N input may promote the dissimilatory reduction of nitrate in the yellow soil and thus help to mitigate MeHg production (Fig. S4, $p < 0.05$). Differently, total C and N in the black soil were approximately three times higher than those in the yellow soil, but the dissolved inorganic N in the black soil was quite low (Table S1). This suggests that available N (e.g., NH_4^+) is limited in this soil despite of nitrate addition. Therefore, the effect of stoichiometric ratios of C/N/S on the NH_4^+ in the black soil was negligible (Fig. S3B). The inhibitory effect might be related to the negative correlations between soil MeHg and multiple denitrifying gene abundances, e.g., *nirS*, *nirK*, and *narG* (Fig. S5A-C, $p < 0.05$), suggesting potential substrate competition between Hg methylating and denitrifying communities. The changes in soil-water stoichiometric ratio may also explain previous observations that the co-application of biochar and nitrate can reduce MeHg production [66]. Consequently, we propose that added N can mitigate C promoting effect on MeHg production through distinct pathways, and further study based on labeled N isotope may help to unveil the underlying mechanisms.

3.2. Hg microbial methylators responding to C N and S input

Our results showed that C and CN amendments consistently increased Deltaprobacteria-*hgcA* and archaea-*hgcA* abundances in the yellow soil and Deltaprobacteria-*hgcA* abundances in the black soil (Fig. 1C-1D; Fig. S6A-B). Soil MeHg concentration showed a positive correlation with Deltaprobacteria-*hgcA* abundance in the yellow and black soil (Fig. S7A-S7B), while no significant relationships between archaea-*hgcA* and MeHg were observed in the two soils. Despite the limitation of using *hgcA* abundance in predicting soil Hg methylation [34], microbes carrying *hgcA* genes have been evidenced to be useful in identifying potential Hg methylators [23,35,45].

To explore the potential Hg methylators in response to the imbalanced C, N, and S input, we further characterized $hgcA^+$ microbial communities. Results showed that most *hgcA* sequences were close to the genera of *Geobacter*, *Methanoregula, Ruminiclostridium, Synthrophus, Methanocella* in the two soils (Fig. 2A-2B). The relative abundance of *Geobacter* was significantly higher than other $hgcA^+$ genera in both soils, but the proportions of $hgcA^+$ taxa varied with soil stoichiometry of C/N/S. The bloomed *Geobacter* could benefit from more small molecular organic compounds due to the nutrient amendments [50]. A significantly positive correlation between the relative abundance of *Geobacter* and soil MeHg further indicate that *Geobacter* played a crucial role in Hg methylation (Fig. 2C-D). These observations are in line with earlier studies showing the dominant role of *Geobacter* in MeHg formation in anaerobic environments [10,68]. However, added N mitigated the

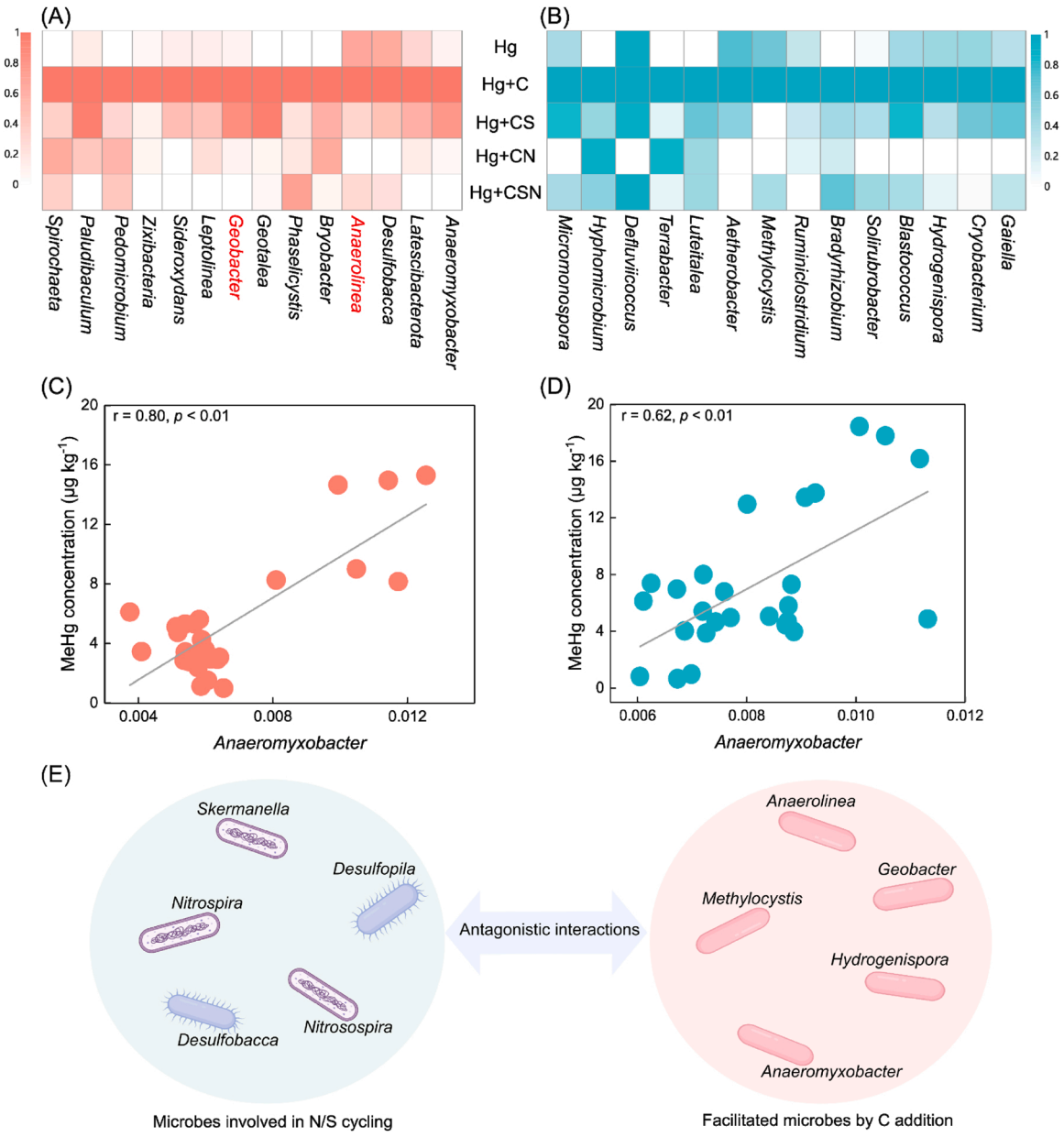


Fig. 3. (A, B) Changes in the relative abundances of the selected soil bacterial families in different treatments of the two soils. (C, D) Relationships between soil MeHg and selected non-Hg methylators (*Anaeromyxobacter*) in the two soils. (E) A conceptual model showing potential antagonistic relationships between microbes involved in N/S cycling and the facilitating taxa by C addition.

C-promoting effect on the proliferation of *Geobacter* in the yellow soil, and this could be an important reason for the decreased MeHg production in the CN and CNS treatments compared to that in the single C treatment. These observations accord with previous studies indicating the relative abundance of *Geobacter* decreased with N addition while increased with C addition in paddy soil [54,55,61]. The inhibitory effect of added N on MeHg accumulation in the two soils might be related to increased proportions of methanogens. For example, the C and N added groups had higher relative abundances of *Methanoregula*, *Methanocella*, and *Methanomassiliicoccus* than those of the C alone treatment, and the methanogens were also reported to have potential to degrade MeHg by oxidizative dementhylation pathway [5].

Different to the effect of N addition, we did not observe great effects of added S on *hgcA*$^{+}$ community compositions in the two soils, suggesting the mitigating effect of S amendment on MeHg production is not attributed to changes in Hg methylating taxa. Previous studies have demonstrated that sulfate input can reduce Hg bioavailability via HgS precipitation in environments, which varied along the concentration of sulfate [37]. In this study, the amount of added S was relatively low, which may favor microbial reduction of sulfate in the soils, leading to declined accessibility of Hg to methylators. However, future study on Hg speciation together with associated microbial communities in soil-water systems would advance our understanding of MeHg production in response to the changes in nutrient element stoichiometry.

3.3. Imbalanced elements input influences MeHg production through microbial interactions

There were significant shifts of overall bacterial community composition after the addition of C, N, and S. The proportions of certain microbial taxa were associated with soil MeHg production in both soils based on the analyses of random forest model ($p < 0.05$) (Fig. S4). For example, the relative abundances of *Anaeromyxobacter*, *Latescibacterota*, and *Anaerolinea* were much higher in the C amended soils comparing

with the control, while significantly decreased after N addition (Fig. 3A). Similar results were also found in the black soil where the relative abundances of *Anaeromyxobacter*, *Gaiella*, *Crybacterium*, and *Hydrogenispora* were reduced by added N or S (Fig. 3B). In particular, we found significant and positive relationships between the proportion of *Anaeromyxobacter* and MeHg content in both soils (Fig. 3C-D), suggesting that this genus can be used as an important biomarker of MeHg prediction in soil. Although some bacteria (e.g., *Syntrophobacter*) have been confirmed to contribute to Hg transformation through potential bacterial syntrophy [59], facilitated microbes may also antagonize Hg methylators and thus inhibit MeHg production in environments. As is well known, however, soil microbes are susceptible to changes in environmental condition. Our results showed that addition of N or S stimulated the proportions of multiple anaerobic taxa involved in the cycling of elements such as C, N and S. These active microbes may compete for available substrates and result in antagonisms between the soil processes driven by these microbial species (Fig. 3E). For example, previous studies suggest significantly antagonistic potential of the taxa such as *Nitrosopira* and *Desulfopila* (both are also potential Hg methylators) involved in N and S cycling with other C-facilitated taxa including Hg methylators [1,47]. This is in line with our observations that single addition of C, N, and S enhanced soil MeHg production, suggesting their important roles in facilitating soil microbes associated with soil Hg methylation. However, simultaneous amendment of these elements also triggered microbial process of multiple biogeochemistry in addition to Hg methylation. Therefore, the enhanced microbial antagonisms by adding N and S in the soils may contribute to the reduced C-promoting effect on MeHg production.

4. Conclusions

Element stoichiometric ratios are key indicators of resource limitation for organisms in environments, and the imbalance of stoichiometry is known to have significant effects on biogeochemical cycles [11]. However, little is known on the effects of imbalanced nutrient elements input through fertilization on microbial-driven formation of neurotoxic MeHg that is posing health risks to humans. Our study provides empirical evidence that certain stoichiometric ratios of C/N/S are crucial for MeHg production in waterlogged paddy soils, which has important implications for our understanding MeHg accumulation in rice paddies. We report that C input alone markedly stimulate MeHg production in typical paddy soils, while S or N inputs can mitigate this promoting effect to some extent. Agricultural activities such as fertilizer application, straw return and sewage irrigation may disproportionately result in increases of soil C and other nutrients [20], which can influence multiple biogeochemical processes including Hg transformation. In addition, projected intensive effect of global changes (e.g., warming, N deposition and acid rain) on element cycling in the future could complicate the scenario occurs in farmlands [4], obscuring our initial recognition on MeHg production in paddy soil. Therefore, our findings may help to improve the prediction of MeHg accumulation in farmlands based on element stoichiometric attributes. Moreover, our work may provide an explanation to different effects of biochar from many sources on the MeHg production [6,7].

Our results highlight the regulation of C, N, and S input on MeHg production through shifting Hg methylating communities. In particular, the addition of N inhibited the key Hg methylators such as *Geobacter*, which has been known play crucial roles in driving MeHg production in various environments [10,33]. Amendments of N or S can also mitigate the C-promoting effect on MeHg production by blooming biogeochemical cycling themself, which may antagonize Hg methylation process by microbes in the soils. These findings provide mechanistic understanding of element stoichiometric effects on MeHg production in paddy soils. Importantly, our work paves an avenue to potential remediation of MeHg pollution in farmlands through optimizing fertilization practices, which is fundamental to mitigations of Hg risk in rice grains relating to billions of people through staple food consumption. Our findings may also have implications for understanding Hg transformation in other aquatic ecosystems such as wetlands, marine, and river sediments, as well as the relationship between humification of organic matter and MeHg production due to the dynamic changes of C/N ratios in the soil [8]. However, we note that the present study has limitations in quantifying relationships between element stoichiometric ratios and MeHg production, which may also be affected by other geochemical factors such as the composition of organic matter, N, and S speciation. We also suggest that further studies linking Hg transformation and multiple microbial processes involved in the cycling of C, N, and S would improve our predictions of MeHg accumulation and environmental risks in water-logged habitats.

Environmental implications

The imbalance of soil nutrient elements is known to have significant effects on biogeochemical cycles. However, little is known on the impacts of combined input of some key nutrient elements on microbial-driven formation of methylmercury that is posing health risks to humans. Our study provides empirical evidence that amendments of N and S can mitigate the C-promoting effect on MeHg production by regulating certain soil microbiota. This study has important implications for better understanding of microbes-driven Hg conversion in paddies and wetlands with nutrient elements input.

Associated content

All data needed to evaluate the conclusions in the paper are present in the paper and/or the Supporting Materials. This material is available free of charge via the Internet at http://xxxxxx.

Conflict of interest

The author declares no conflict of interest.

Data Availability

Data will be made available on request.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jhazmat.2023.130983.

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